Latency-Aware Phase Optimization for Reconfigurable Intelligent Surfaces Serving Mobile Users

Seda Savaşçı Şen, Ali Çalhan, and Murtaza Cicioğlu

***Abstract*—A walking user changes the radio channel within tens of milliseconds, yet reconfigurable intelligent surface (RIS) phase optimizers are usually compared by the rate they reach on the channel they were given. We instead evaluate each method on the channel that exists when it finishes, its *effective rate*, with measured latencies, in a 3GPP indoor-office multi-user MISO downlink with a 32-element RIS. First, coordinate descent and long-run Adam, the methods with the highest rate at $t = 0$, need 0.9–5.9 s and lose to a 2.5 ms single-user heuristic. Second, more iterations help only up to a point: at 1.5 m/s Adam's effective rate peaks at 4–6 steps and running it to 48 steps costs 14.0%; at 10 km/h the peak is at 3 steps. The optimum is set by the step time relative to the Doppler frequency, and a small-delay model explains how it scales. Third, a learned update rule that beats tuned Adam at equal steps is dominated by faster Adam configurations once latency is measured on one CPU thread. All results assume the precoder is recomputed from the current channel. For pedestrian users at the measured step cost, latency, not convergence, decides which method is best.**

***Index Terms*—Reconfigurable intelligent surface, mobility, computation latency, effective rate, channel aging, deep unfolding, multi-user MISO.**

I. Introduction

A reconfigurable intelligent surface (RIS) can turn a blocked corner of a building into a covered one [1]–[4], and prototypes have shown that it works [5], [6]. But it works only if its passive elements steer the signal toward where the user *is*, not toward where the user *was*. Choosing those phases is a hard, non-convex problem, attacked with iterative solvers [2], [7]–[10] and, increasingly, with learned policies [11]. These methods are usually compared by the rate they reach on the channel they were given.

That comparison hides a race. A person walking through an office changes the radio channel within tens of milliseconds; a careful solver may need seconds. By the time the best answer is ready, the question has changed. A rough phase delivered on time can then beat an excellent phase delivered late, and running more iterations can make the result worse, not better.

This paper measures that race. We evaluate every method on the channel that exists when it finishes, its *effective rate*, with latencies measured on a CPU, in the 3GPP indoor-office scenario with walking users [12]. The question is the one a base station (BS) faces: once time is counted, which method should it run, and for how long?

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A. Contributions

1) **A latency-aware criterion.** We define the effective rate (Section IV) and show that under it the solvers with the best rate on paper lose to a single-user heuristic that takes milliseconds, for pedestrian users at the measured step cost.
2) **An optimal number of iterations.** More iterations help only up to a point. The best number of steps is set by a single quantity, the time per step relative to the Doppler frequency, and a small-delay model explains how it scales. The optimum persists for a second classical optimizer, a larger surface and every standardized indoor user speed.
3) **Periodic operation.** When the BS re-optimizes continuously, the pilot overhead decides how often to update, and each update should still be short.
4) **Learning on the clock.** A learned update rule that beats a tuned classical optimizer step for step loses to it millisecond for millisecond.
5) **The price of the main assumption.** All results assume that the BS refreshes its precoder from the current channel; we quantify how much the results depend on it.

II. Related Work

RIS phase optimization. Joint precoding and phase design has been addressed with alternating optimization and semidefinite relaxation [2], practical phase-dependent amplitude models [7], block coordinate descent [8], fractional programming [9], fixed-point and manifold methods [13] and weighted minimum mean-square error (WMMSE) methods [10], [14]. These works report the objective value reached; run time, when discussed, is a complexity order, not a latency measured against the coherence time.

Learning. Learned policies include deep reinforcement learning for joint precoding and phase design [11] and learning to reflect directly from pilots [15]. Algorithm unrolling turns an optimizer's iterations into network layers [16]; it has been applied to RIS beamforming [17], MIMO detection [18] and WMMSE precoding [19], [20], and trained networks can

approximate WMMSE at a fraction of its run time [21]. Phase-aware regression, which predicts the sine and cosine of each phase, removes the $\pm\pi$ discontinuity of direct phase regression [22]. Learned methods are usually motivated by speed, but that speed is rarely related to how fast the channel changes, and classical baselines are rarely tuned or timed under the same conditions.

Mobility and outdated channels. Mounting the surface on a vehicle turns fast fading into slow fading [23]. Channel aging [24] and outdated channel knowledge [25] have been analyzed for RIS, predictive beamforming forecasts the channel of fast unmanned aerial vehicle (UAV) links [26], and active sensing tracks mobile users [27]. Aging is classical in massive MIMO [28], [29] and broadcast channels [30]. Two-timescale designs update the phases from channel statistics and the precoder from instantaneous channels [31], [32]. Practical surfaces have a few bits of phase resolution [33], [34]; we use continuous phases. In all of these works the delay between channel acquisition and use is a fixed system parameter; the optimizer's own run time, which differs from method to method and from one iteration count to the next, is not part of the model.

Gap. To our knowledge, no RIS work evaluates each method on the channel at its own measured completion time, shows that this creates an optimal iteration count, or compares learned and classical optimizers both at equal steps and at equal measured time.

III. System and Channel Model

A. Signal Model

A BS with a uniform linear array of $N_t = 8$ half-wavelength-spaced antennas serves $K = 4$ single-antenna users through a direct link and an RIS of $N = 32$ passive elements (an 8×4 planar array); Section VI-C also uses $N = 64$. With phases $\boldsymbol{\theta} \in [0, 2\pi)^N$, $\mathbf{v} = e^{j\boldsymbol{\theta}}$ and precoder $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_K]$, user k receives

$$y_k = \big(\mathbf{h}_{d,k}^H + \mathbf{g}_k^H \,\mathrm{diag}(\mathbf{v})\mathbf{F}\big) \sum_{j=1}^{K} \mathbf{w}_j s_j + n_k, \qquad (1)$$

where $\mathbf{F} \in \mathbb{C}^{N \times N_t}$ is the BS–RIS channel, $\mathbf{g}_k^H$ the RIS–user channel, $\mathbf{h}_{d,k}^H$ the direct channel and $n_k \sim \mathcal{CN}(0, \sigma^2)$. With $\mathbf{h}_k^H(\boldsymbol{\theta})$ the bracketed effective channel, user k has rate $R_k = \log_2(1 + \mathrm{SINR}_k)$, with signal-to-interference-plus-noise ratio $\mathrm{SINR}_k = |\mathbf{h}_k^H \mathbf{w}_k|^2 / (\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \sigma^2)$. The precoder is regularized zero-forcing [35] on the effective channel, with regularizer $\lambda = K\sigma^2/P$ and total power P. The objective is the minimum user rate, a fairness criterion also used for RIS-assisted multiple-input single-output (MISO) systems in [36],

$$\max_{\boldsymbol{\theta}} \; \min_k R_k(\boldsymbol{\theta}), \qquad (2)$$

which Adam, Armijo and the learned update (Section V) optimize through the smooth surrogate $-\frac{1}{\beta}\log\sum_k e^{-\beta R_k}$, $\beta = 8$; coordinate descent and the long-run Adam reference maximize the hard minimum.

B. Geometry and Propagation

The room is $40 \times 20 \times 6$ m (Fig. 1). The RIS is centered at $(20, 0.1, 3)$ m on a wall; the BS is at $(20, 6, 4)$ m, in the panel's boresight and beyond its Fraunhofer distance. Users are dropped uniformly at heights of 1–2 m. The carrier is $f_c = 3$ GHz, the bandwidth 20 MHz, the transmit power 20 dBm, the BS antenna gain 5 dBi and the noise figure 7 dB. All links follow the 3GPP TR 38.901 indoor-office model [12]: its path loss, line-of-sight (LoS) probability, Rician factor and log-normal shadowing with Gudmundson correlation [37] at 5 m (TR 38.901 specifies 10 m for LoS and 6 m for non-LoS). Non-LoS components come from point scatterers shared by all links of a scenario. The cascaded amplitude follows the two-hop model of [38] with a $\cos^q$ element pattern. Because an RIS is deployed where the direct path is obstructed, the direct link carries an extra 40 dB of loss.

C. Mobility

Users walk in straight lines at $v = 1.5$ m/s (5.4 km/h) and are reflected off the walls. This lies between the 3 km/h of the 3GPP indoor-hotspot evaluation [39] and the 10 km/h upper end of the IMT-2020 pedestrian class, at which indoor mobility is evaluated [40]; Section VI-D repeats the main measurements at these and higher speeds. Scatterers, path gains, LoS states, shadowing and $\mathbf{F}$ are held fixed; only the users move, and $\mathbf{g}_k(t)$ and $\mathbf{h}_{d,k}(t)$ are recomputed from geometry at every instant. The reference time scale is the Doppler frequency $f_d = v f_c / c = 15$ Hz, with coherence time $T_c = 0.423/f_d = 28.2$ ms [41], [42].

Limitations. Fixed scatterers make the channel change less than a real one over seconds, so effective rates beyond 200 ms are optimistic for slow methods; the true values should lie between them and the stale floor of Table I. Scattered-path gains are drawn independently per BS antenna, which makes all absolute rates optimistic; Section VI-D repeats the main measurements with shared gains. The effect of fixed scatterers on the optimal step count is not known. Channel knowledge is perfect apart from the acquisition delay we add. The model is narrowband (no beam squint), and far-field for $N = 32$; for $N = 64$ the BS (6.0 m away) lies just inside the 6.4 m Fraunhofer distance, and the channel uses exact per-element distances.

IV. Latency-Aware Evaluation Framework

A. Effective Rate

Let a method A receive the channel $\mathcal{H}(0) = \{\mathbf{F}, \mathbf{g}_k(0), \mathbf{h}_{d,k}(0)\}$, return the phase $\boldsymbol{\theta}_A$, and need the wall-clock time τ_A to do so.

Definition 1 (Effective rate). The effective rate of A is its minimum user rate on the channel that exists when it finishes,

$$R_A^{\mathrm{eff}} = \min_k R_k\big(\boldsymbol{\theta}_A;\, \mathcal{H}(\tau_A)\big), \qquad (3)$$

with the precoder recomputed from $\mathcal{H}(\tau_A)$.

The precoder is a closed-form computation that the BS repeats whenever it has fresh pilots; the RIS phase needs the iterative optimization whose latency we measure. Definition 1

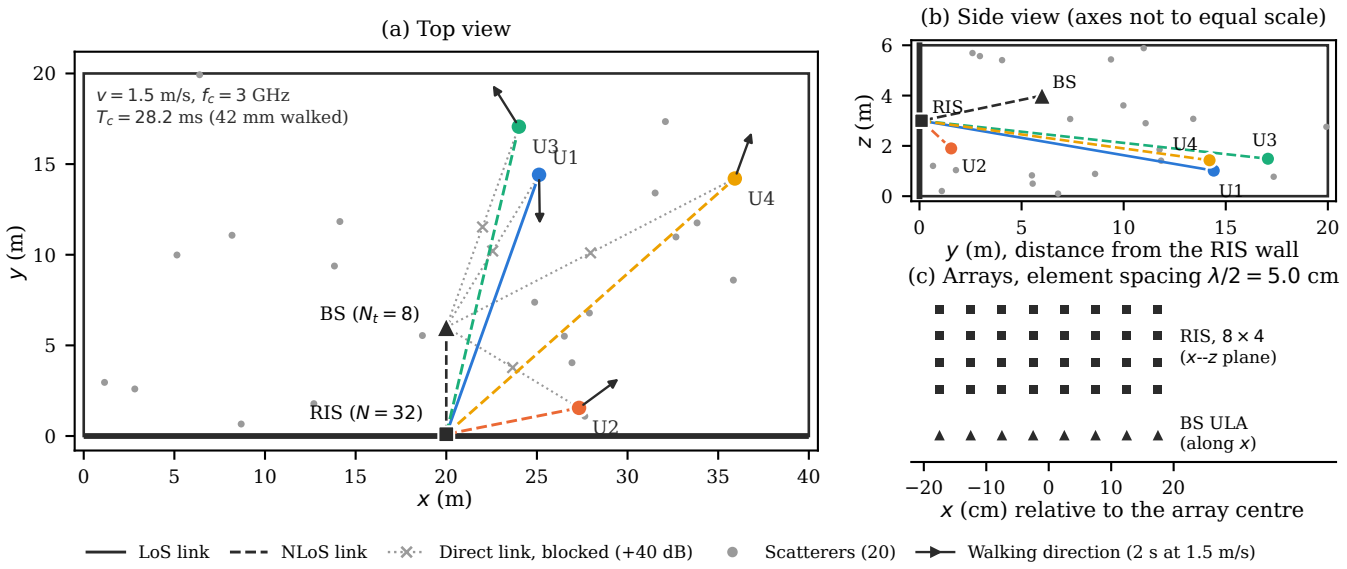


Fig. 1. The simulated scenario, showing the first scenario of the aging set as produced by the channel generator. (a) Top view of the $40 \times 20 \times 6$ m office: BS at $(20, 6, 4)$ m, wall-mounted RIS centered at $(20, 0.1, 3)$ m, four users walking at 1.5 m/s, 20 point scatterers. Line style: line-of-sight (LoS) state of each link. All direct links carry the 40 dB blockage. Arrows: 2 s of walking; within one coherence time a user moves only 42 mm. (b) Side view. (c) Element layouts.

therefore lets the phase age and keeps the beam fresh, a split similar to that of two-timescale designs [31]. Section VI-G prices this assumption.

B. Why an Optimal Number of Steps Exists

For an iterative method run for L steps, write $r_0(L)$ for its minimum rate at $t = 0$, $\tau(L)$ for its latency, and $\Delta(L) = r_0(L) - \min_k R_k(\boldsymbol{\theta}_L; \mathcal{H}(\tau(L)))$ for the rate lost to aging during its own run time, so that $R^{\text{eff}}(L) = r_0(L) - \Delta(L)$. Each step raises r_0 with diminishing returns, while Δ grows with τ. A step pays off only while

$$r_0(L+1) - r_0(L) \;>\; \Delta(L+1) - \Delta(L), \tag{4}$$

and the effective rate peaks where this fails: the *intrinsic optimum* $L^\star$. It depends jointly on the optimizer, its per-step latency and the rate of channel variation. Whether R^{eff} is unimodal is an empirical question (Section VI-B).

C. Equal Steps versus Equal Milliseconds

If one learned step costs as much time as c classical steps, an equal-step comparison gives the learned method c times the time budget. We therefore also compare at equal measured time.

Definition 2 (Dominance). Configuration B dominates A if $\tau_B \le \tau_A$ and $r_{0,B} \ge r_{0,A}$, with one inequality strict.

A dominated configuration is never the better choice at any latency budget.

D. Normalized Step Cost and a Small-Delay Model

Channel variation depends on the distance a user moves in wavelengths, $f_d\tau$. If an optimizer takes ΔT per step, its latency after L steps in these units is Lx, with

$$x = \Delta T\, f_d. \tag{5}$$

In our model, scaling ΔT is exactly equivalent to scaling the speed v; a change of carrier is only approximately equivalent, since the path loss and the array size in wavelengths change with it.

Two approximate regularities, checked in Section VI-C, make (4) explicit. The convergence gap decays roughly as a power law,

$$R_\infty - r_0(L) \approx D\, L^{-\gamma}, \tag{6}$$

and for small delays the aging loss is quadratic in the distance moved, $u = f_d\tau$,

$$\Delta(L, u) \approx \kappa_L\, u^2, \tag{7}$$

as in the leading term of the Clarke correlation, $J_0(2\pi u) \approx 1 - \pi^2 u^2$ [41], [42]. With $u = xL$,

$$R^{\text{eff}}(L) \approx R_\infty - D\, L^{-\gamma} - \kappa\, x^2 L^2. \tag{8}$$

Proposition 1 (Scaling of the optimum). For $\gamma, D, \kappa, x > 0$ and $\kappa_L = \kappa$, (8) has a unique maximizer over $L > 0$,

$$L^\star = \left(\frac{\gamma D}{2\kappa}\right)^{\frac{1}{\gamma+2}} x^{-\frac{2}{\gamma+2}}, \tag{9}$$

at which the aging loss equals $\gamma/2$ times the remaining convergence gap.

Proof. The derivative $\gamma D L^{-\gamma-1} - 2\kappa x^2 L$ is positive for small L, negative for large L, and vanishes only at $L^{\gamma+2} = \gamma D/(2\kappa x^2)$; the second derivative is negative. Substituting gives the balance. □

The optimum is thus a power law in x with exponent in $(-1, 0)$: a faster optimizer should run more steps, but less than proportionally more. The derivation assumes $xL^\star \ll 1$, which holds over our range ($xL^\star \le 0.16$). With fitted constants, (9) describes the measurements rather than predicting them independently; its value lies in explaining how the optimum shifts.

E. Periodic Re-optimization

A deployed controller repeats the optimization with period U: it spends T_{p} on pilots, without data, then $L\Delta T$ on optimization, deploys the result and waits ($U \geq T_{\mathrm{p}} + L\Delta T$; *back-to-back* means equality).

Definition 3 (Time-averaged effective rate). With $\boldsymbol{\theta}(t)$ the deployed phase and $\mathcal{P}$ the pilot intervals, over a window $\mathcal{W}$

$$\bar{R}^{\mathrm{eff}} = \frac{1}{|\mathcal{W}|} \int_{\mathcal{W}\setminus\mathcal{P}} \min_k R_k\big(\boldsymbol{\theta}(t);\, \mathcal{H}(t)\big)\, dt. \tag{10}$$

Each update starts from scratch (*cold*) or from the deployed phase (*warm*).

V. Measurement Protocol

A. Scenario Sets

All sets use seed 11 unless stated otherwise, and numbers from different sets are never mixed in one comparison.

- *Aging set* (Tables I–III, Figs. 2–4): 400 scenarios with walking users and 20 point scatterers, close to the 15–19 clusters of the 3GPP indoor-office model; with $N = 64$, the same construction with an 8×8 panel.
- *Beam set* (Section VI-G): 400 further scenarios with 6 scatterers.
- *Static set* (Fig. 5): 6000 scenarios without mobility, of which 4200 are used for training, 400 for validation and 600 for testing.

The generator is not prefix-stable, so every trained model is evaluated with the sample count it was trained with.

B. Methods

SU-average and *SU-best-of* are closed-form single-user phase alignments; the latter keeps the best of $K+1$ candidates. *Adam* [43] optimizes the soft-min surrogate, starting from SU-average. *Armijo* is steepest ascent with Armijo's rule [44] on the same surrogate, with a per-scenario backtracking line search. *Coordinate descent* searches each element over 16 phase levels, for up to 8 sweeps; *long-run Adam* runs 1000 iterations from 3 random starts. The *learned update* is a deep-unfolded network with L learned steps that takes the gradient of the soft-min rate as input and exchanges information across users; one network is trained per L, without labels.

Tuning. On the static set, Adam's step size and objective (soft-min or hard min) are selected on the validation set at every budget: from ten values in $[0.02, 2.0]$ for $L \leq 16$, where 0.75 with the soft-min, an interior point, is selected every time, and from six values in $[0.2, 1.5]$ for $L = 24$–64. On the aging set Adam uses step size 0.5 (0.25, 0.75 and 1.0 give lower peaks), and Armijo's initial step is chosen from five values (3 is selected); both choices use the reported scenarios, so they are robustness checks rather than held-out selections.

C. Timing

Latencies are wall-clock times for one scenario on an Intel Xeon CPU at 2.1 GHz with one PyTorch thread; the median over repetitions is kept, and most measurements are repeated in several rounds. These are latencies of unoptimized Python code; Section VI-C shows what changes for faster code. The Adam rows use 13.3/8 = 1.66 ms per step, from an earlier two-thread measurement made before a precoder fix; this value lies within the 1.49–2.02 ms measured here. On the aging set the load average was 0.20 before and 0.86 after timing (the rise is the timing script's own thread), and the round-to-round spread was 1.15–3.66×. Armijo costs 1.26 times as much as Adam per step (2.09 ms; per-scenario ratio 0.81–1.97). On the static set the spread was 1.15–1.60×, the load average was not recorded, and the learned network's input features are excluded from its time, which favors it. The $N = 64$ step-time ratio of 1.04 comes from one round on 20 scenarios. Coordinate descent with 1–8 sweeps is timed in three rounds from a load average of 0.45.

D. Statistics and Pilots

Effective rates are interpolated on a time grid at each method's latency plus an optional acquisition delay T_{ov}. Because absolute rates vary strongly across scenarios ($\pm$0.17–0.24 b/s/Hz for the means of Table I), methods are compared with paired bootstrap confidence intervals (CIs) [45] over scenarios (2000 resamples). Differences of a few thousandths of a b/s/Hz between neighboring L are not resolved. Best configurations are selected and reported on the same scenarios unless stated, without a multiple-comparison correction, and should be read as estimates.

We tie T_{ov} and the pilot time T_{p} to cascaded channel estimation [46]. With the scheme of [47], 48 pilot symbols suffice for $K = 4$, $N = 32$, $N_t = 8$. With 5G New Radio (NR) numerology [48] and one orthogonal frequency-division multiplexing (OFDM) symbol per pilot they last 3.43, 1.71 and 0.86 ms at 15, 30 and 60 kHz subcarrier spacing; feedback, estimation processing and RIS switching come on top and are not modeled. The periodic controller is simulated on a 0.5 ms grid over 200 ms, with update periods of 10–80 ms or back-to-back, averaging over 100–200 ms.

Reproducibility. One script generates every table and figure from stored result files; rerunning the aging sweeps reproduces them bit for bit. A separate audit script checks 123 assertions about the learning experiments of the underlying thesis; among the results here it covers those of Section VI-D.

VI. Results

A. Slow, Strong Solvers Lose to a Fast Heuristic

In this scenario the RIS carries most of the signal: with the direct links blocked, the median per-user signal-to-noise ratio (SNR) is 20.0 dB for the direct link alone and 31.1 dB through the RIS. Without the RIS the minimum user rate is 1.744 b/s/Hz; a random phase gives 4.072.

Fig. 2 fixes each method's phase at $t = 0$ and follows the minimum rate as users walk. Rates hold for a few milliseconds, drop sharply around T_c and level off above the stale floor. The better a phase was optimized, the more it loses: by T_c long-run Adam falls from 7.117 to 5.473 b/s/Hz, while SU-average only moves from 4.624 to 4.503.

Table I evaluates each method at its own latency. Long-run Adam and coordinate descent have the best $t = 0$ rates

TABLE I
RATE AT $t = 0$, LATENCY AND EFFECTIVE RATE (AGING SET, 400 SCENARIOS, 1.5 M/S). STALE FLOOR: $t = 0$ PHASE ON AN INDEPENDENT CHANNEL. †BEYOND THE 200 MS RELIABLE HORIZON; OPTIMISTIC. *VALUE AT THE LAST GRID POINT (5.6 S).

Method	t=0 (b/s/Hz)	Latency (ms)	Effective (b/s/Hz)	Stale (b/s/Hz)
No RIS (direct link)	1.744	–	–	–
Random phase	4.072	–	–	–
SU-average	4.624	0.42	4.623	4.315
SU-best-of	5.276	2.5	5.272	4.336
Adam, $L = 8$	6.443	13.3	5.849	4.326
Adam, $L = 12$	6.631	20.0	5.515	4.370
Adam, $L = 16$	6.754	26.6	5.324	4.381
Adam, $L = 20$	6.835	33.2	5.276	4.364
Coord. descent	6.925	880.8	4.975†	4.352
Long-run Adam	7.117	5851.7	4.667†*	4.423

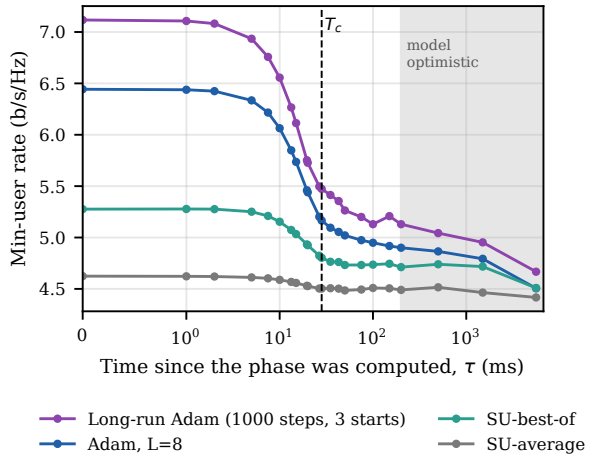


Fig. 2. Minimum user rate of a phase fixed at $t = 0$ as users walk at 1.5 m/s, with the precoder refreshed. Dashed: $T_c = 28.2$ ms. Shaded: beyond 200 ms the model is optimistic.

of the table, but they need 5.9 s and 0.88 s, respectively. By then users have moved up to several meters, and even their optimistic effective rates fall below that of SU-best-of, which takes 2.5 ms: SU-best-of is ahead of coordinate descent by 0.297 b/s/Hz (95% CI 0.204–0.390) and of long-run Adam by 0.604 (0.415–0.805). This is not an artifact of a fixed budget: one multi-user coordinate sweep already takes 107 ms, about four coherence times, and the best of one to eight sweeps (5.014 b/s/Hz, two sweeps) stays 0.26 b/s/Hz below SU-best-of (CI 0.17–0.36).

B. More Iterations Stop Helping After a Few Steps

Fig. 3 sweeps the number of Adam steps from SU-average (4.624 b/s/Hz). The $t = 0$ rate rises with every step, from 5.412 at $L = 1$ to 7.037 b/s/Hz at $L = 48$. The effective rate does not: it peaks at $L = 4$–6 (5.980, 5.982 and 5.970 b/s/Hz, at 6.7–10.0 ms), and every other step count is significantly lower. At $L = 48$ it is 14.0% (95% CI 12.5–15.5%) below the peak, although the $t = 0$ rate there is 13.2% higher. In terms of (4), going from $L = 4$ to 6 adds 0.188 b/s/Hz at $t = 0$ but costs 0.198 in aging.

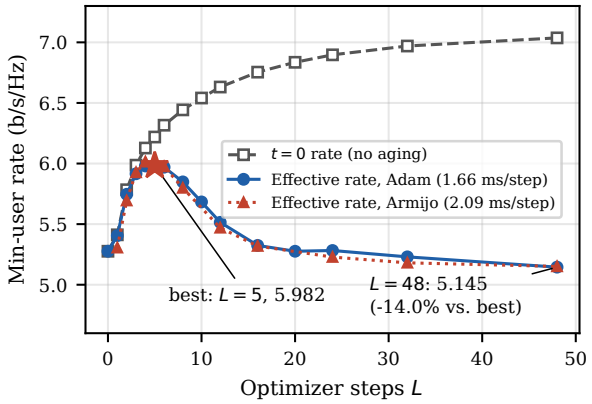


Fig. 3. Step sweep (aging set). The $t = 0$ rate increases with L; the effective rate peaks at $L = 4$–6 and is 14.0% lower at $L = 48$. Armijo (2.09 ms per step) shows the same shape. $L = 0$: SU-best-of at 0.05 ms (5.276; 5.272 at its measured 2.5 ms).

The optimum is not an artifact of Adam or its step size. Step sizes 0.25, 0.75 and 1.0 put it at $L = 6$, 4 and 5, with losses of 11.5%, 13.7% and 11.3% at $L = 48$. Armijo steepest ascent peaks at $L = 4$ (6.015 b/s/Hz). It gains more than Adam in the first steps but costs 1.26 times as much per step, and the two peaks are statistically tied (difference 0.033, CI -0.001 to 0.064). With Armijo's per-step cost at 1.0, 1.2, 1.4, 1.6 and 2.0 times Adam's, its best effective rate is $+0.147$, $+0.050$, -0.013, -0.079 and -0.197 b/s/Hz relative to Adam's: even between two classical optimizers, latency decides the ranking.

C. One Number Sets the Optimum

An optimized implementation would be faster per step. Fig. 4 therefore scales Adam's step time from 0.01 to 3 times the measured value, i.e., x from 2.5×10^{-4} to 7.5×10^{-2}; this setup sits at $x = 2.5 \times 10^{-2}$, and the same axis covers users up to 100 times slower or 3 times faster than here. The best step count falls steadily: 96 steps at the smallest x, 48, 24–32, 12–16, 4–6 here, and 2 at the largest. At the smallest x convergence is nearly free (running to $L = 256$ costs 1.5%, CI 1.3–1.8%); the cost of convergence peaks at about 21% near $x \approx 2.5 \times 10^{-3}$. For $x \geq 2.5 \times 10^{-2}$, $L = 256$ lies beyond the 200 ms horizon, so its cost is, if anything, understated. Whether convergence hurts is therefore a property of the regime, not of RIS optimization in general. Faster hardware, such as a GPU or an NPU, shortens the step and so moves the optimum, but does not remove it: in the model (Proposition 1) an interior optimum exists for every $x > 0$, and in the measurements it persists down to a hundredfold faster step, where it lies at 96 steps.

Both laws behind Proposition 1 hold approximately. The aging loss is close to quadratic ($R^2 \geq 0.98$ for every L, fitted for $\tau \leq 10$ ms), with κ_L growing from 7.8 at $L = 1$ to 25.1 at $L = 48$. The convergence gap roughly follows a power law, with $\gamma = 0.93$, but between 0.56 and 1.19 depending on the fit range, so the predicted exponent lies between -0.78 and -0.63. The measured slope of $L^\star(x)$ is -0.675 (95% CI -0.692 to -0.655), inside that range; the model thus pins the

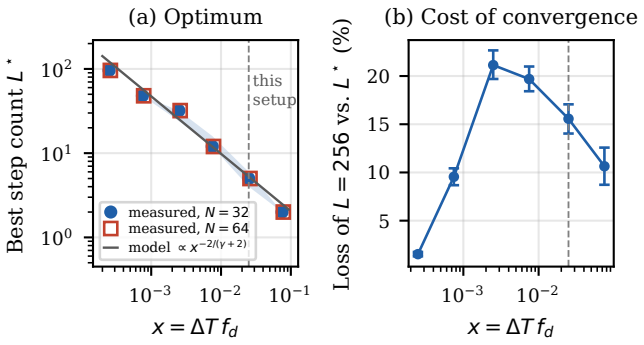


Fig. 4. Adam with its step time scaled from 0.01 to 3 times the measured value. (a) Best step count for $N = 32$ (shaded: not significantly below the best) and $N = 64$; line: closed form (9). (b) Loss of running to $L = 256$ instead of $L^\star$ ($N = 32$), with 95% paired CIs. Dashed: this setup.

exponent down only to about ± 0.08. The closed form (9) is within a factor of 1.27 of the measured optimum at every x (5.3 against 5 here). This is an in-sample check: κ_L is fitted over the delays (1.6–10 ms) at which the optima occur. The balance of Proposition 1 does not hold quantitatively (the ratio varies from 0.19 to 1.65), so the model explains where the optimum lies and why it moves, not how the two losses split.

A larger 8×8 surface ($N = 64$) raises the converged rate (8.48 against 7.10 b/s/Hz) but has the same best step count at every step-time multiplier, which confirms the optimum but not, independently, the exponent. Because its step costs only 1.04 times as much in our code, it sits at nearly the same x; an implementation whose cost grows with N would move it to fewer steps.

D. Standardized Speeds and Acquisition Delay

Table II repeats the main measurements on the same scenarios, on a time grid with 0.5 ms spacing up to 40 ms, at the user speeds of the standards: 3 km/h, used in the 3GPP indoor-hotspot evaluation [39]; 10 km/h, the upper end of the IMT-2020 pedestrian class at which indoor mobility is evaluated [40]; and, beyond the pedestrian class, 18 km/h (5 m/s) and the 30 km/h of the dense-urban evaluation. Without acquisition delay, both findings—an optimal step count and the loss of the slow solvers—hold at every speed. The optimum moves from 8 steps at 3 km/h to 3 at 10 km/h, 2 at 18 km/h and 1 at 30 km/h. Running to $L = 48$ costs 14.5% (95% CI 13.1–16.1%) at 3 km/h and 10.5% (9.0–11.9%) at 10 km/h. SU-best-of is ahead of coordinate descent by 0.25–0.42 b/s/Hz and of long-run Adam by 0.45–0.69 at every speed, with paired CIs above zero (optimistic values for the slow solvers). At 18 and 30 km/h, $L = 48$ lies beyond the reliable horizon (60 and 36 ms), so that cost is approximate. With 3.43 ms of acquisition delay the optimum is not resolved at 18 and 30 km/h, and at 30 km/h SU-best-of and coordinate descent are no longer significantly different. On this finer grid, $L = 4$–6 are within 0.5% of each other at 1.5 m/s, and running to $L = 48$ costs 14.2% (14.0% in the main sweep).

A deployed system also spends time on pilots and feedback before it optimizes. On the main grid, adding an acquisition delay of 0–10 ms at 1–3 m/s moves the optimum the way (4) predicts (at 1.5 m/s from $L = 4$–6 to $L = 3$–4); excluding delays above $T_c/2$ (only 3 m/s with 10 ms), running to $L = 48$ costs 2.9–14.7%. The optimum also tolerates a wrong speed assumption: choosing L for 1.5 m/s when users move at 1.8 m/s costs less than 0.5%, and no pair of assumed and true speeds in our grid costs more than 5.7%.

Shared scattering gains. With one gain per scatterer for all BS antennas, as in a physical channel, absolute rates fall (Adam's best effective rate at 1.5 m/s on the grid of Table II: 5.398 instead of 5.984 b/s/Hz), but the findings hold: $L^\star$ stays at 4–6 at 1.5 m/s and 3 at 10 km/h, running to $L = 48$ costs 17.7% (CI 15.8–19.5%) and 14.2%, respectively, and SU-best-of stays ahead of coordinate descent and long-run Adam (paired CIs above zero).

TABLE II
STANDARDIZED AND HIGHER USER SPEEDS (AGING SET, NO ACQUISITION DELAY): COHERENCE TIME, NORMALIZED STEP COST, BEST STEP COUNT (RANGE WITHIN 0.5%), COST OF RUNNING ADAM TO $L = 48$, AND Δ: SU-BEST-OF MINUS COORDINATE DESCENT IN EFFECTIVE RATE (B/S/HZ). COMPUTED ON A 0.5 MS GRID, SO VALUES CAN DIFFER SLIGHTLY FROM TABLE I AND FIG. 3. †BEYOND THE RELIABLE HORIZON; APPROXIMATE.

km/h	Source	T_c (ms)	x	$L^\star$	L=48	Δ
3	3GPP [39]	50.8	0.014	8 (6–8)	14.5%	+0.27
5.4	ours	28.2	0.025	4 (4–6)	14.2%	+0.31
10	ITU [40]	15.2	0.046	3	10.5%	+0.25
18	ITU, veh.	8.5	0.083	2	8.0%†	+0.42
30	ITU, urban	5.1	0.139	1	4.3%†	+0.42

TABLE III
PERIODIC RE-OPTIMIZATION (AGING SET): BEST UPDATE PERIOD U (MS; B2B = BACK-TO-BACK) AND STEPS PER UPDATE L, WITH THE TIME-AVERAGED EFFECTIVE RATE (B/S/HZ) FOR PILOT DURATIONS T_p. WARM−COLD: PAIRED DIFFERENCE OF THE TWO BEST POLICIES; §NOT SIGNIFICANT.

	Cold start			Warm start			
T_p (ms)	U	L	rate	U	L	rate	warm−cold
0	b2b	3	5.83	b2b	2	6.36	+0.52
0.86	20	4	5.20	10	3	5.42	+0.22
1.71	b2b	48	5.09	20	3	5.23	+0.15
3.43	b2b	48	4.98	80	8	4.99	+0.01§

E. Periodic Operation: Pilots Set the Period, Updates Stay Short

Table III simulates the controller of Definition 3. The pilot overhead decides how often to update. With free pilots, back-to-back updates with two warm-started steps are best (6.36 b/s/Hz); with 0.86 ms of pilots the best period is 10–20 ms; with 3.43 ms, updates every 80 ms (warm start) and back-to-back updates with 48 steps (cold start) are statistically tied. What does not change is that short updates suffice: in every row some policy with at most 8 steps is not significantly below the best. Warm starting gains up to 0.52 b/s/Hz when pilots are short and nothing significant with 3.43 ms pilots.

F. A Learned Update Rule Wins per Step but Loses per Millisecond

This comparison uses the static set and $t = 0$ rates, to isolate compute efficiency. At an equal number of steps the learned rule beats tuned Adam at every budget $L \in \{1, 2, 4, 8, 16\}$ (Fig. 5(a)), winning on 68–84% of test scenarios. But one learned step costs as much time as 2.7–4.0 Adam steps. On the measured time axis (Fig. 5(b)) every learned configuration is dominated by an Adam configuration that is both faster and better, in each of five timing rounds. For example, the learned rule with $L = 4$ needs 18.41 ms for 5.285 b/s/Hz, while Adam with $L = 8$ needs 13.48 ms for 5.380 b/s/Hz. The equal-step ranking reverses completely. Aging is not included here; since the learned configuration is the slower one in every pair, aging would be expected to widen the gap. The cost ratio, however, is a property of this implementation on one CPU thread: on hardware that accelerates the network's matrix products more than Adam's step, the reversal could weaken.

G. The Fresh-Beam Assumption Matters

All results assume that the BS refreshes its precoder from the current channel while only the RIS phase ages. On the beam set, keeping the precoder from $t = 0$ instead makes the rate collapse within tens of milliseconds: at T_c it is 11.7–14.2$\times$ lower than with a fresh beam, for every phase choice, and with the phase of Adam ($L = 8$) a precoder only 5 ms old already lowers the rate from 5.228 to 1.736 b/s/Hz (beam-set rates are not comparable with Table I). Refreshing it is cheap (0.12 ms, about 13 times shorter than one Adam step), but it requires effective-channel pilots every few milliseconds, whose airtime we do not charge.

VII. Conclusion

We asked which RIS phase optimizer a base station should run when users move, and for how long. Counting the time each method needs, we find the following.

1) **Speed beats precision.** Coordinate descent and long-run Adam need 0.9–5.9 s and lose to a 2.5 ms single-user heuristic, even with a tuned number of coordinate sweeps; without acquisition delay, this holds at every speed from 3 to 30 km/h (with optimistic values for the slow solvers).
2) **A few steps are optimal.** At 1.5 m/s Adam's effective rate peaks at 4–6 steps; running it to 48 steps costs 14.0%. The optimum falls to 3 steps at 10 km/h, the upper end of the IMT-2020 pedestrian class.
3) **One number sets the optimum.** The best step count is governed by the step time relative to the Doppler frequency, $x = \Delta T f_d$, and scales approximately as $x^{-2/(\gamma+2)}$ (measured exponent -0.675); with a hundredfold faster implementation, convergence becomes nearly free.
4) **Pilots set the update period.** In continuous operation the pilot overhead decides how often to update; each update should still be short.
5) **Learning must win on the clock.** A learned update rule that wins at equal steps (static set, one CPU thread) is dominated by faster classical steps once latency is measured.

In the regime quantified here, latency is therefore not a secondary metric for mobile RIS: it decides which method is best.

Practical rules. Choose the number of steps from x and the acquisition delay, not from convergence. Choose the update period from the pilot overhead and keep each update short. Compare methods on measured latency against a classical baseline tuned at each budget, and report dominance rather than equal-step wins.

Limitations. Fixed scatterers make long-horizon rates optimistic, and independent per-antenna gains make absolute rates optimistic, although the findings hold with shared gains. All effective rates assume that the BS recomputes its precoder from the current channel within a few milliseconds, with effective-channel pilots whose airtime we do not charge. Channel knowledge is perfect apart from the delays we add, RIS switching time is not modeled, and the model is narrowband with one geometry. Latencies come from single-threaded Python. Faster hardware moves the optimum but, as long as a step takes time, does not remove it; the cost ratio of learned and classical steps on GPUs or NPUs remains to be measured. The normalized step cost carries the results to faster code and other speeds, and approximately to other carriers. Long-run Adam was run at a fixed budget, and coordinate descent with one to eight sweeps; semidefinite relaxation [2], WMMSE [10] and manifold methods were not timed. Our next step is wideband OFDM operation, where beam squint makes the best phase depend on frequency and adds a further trade-off between quality and time; imperfect channel knowledge and discrete phases in the multi-user case follow.

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Data and Code Availability

The scripts that generate every table and figure, and the stored measurement files they read, are available at https://github.com/sedasavascisen/ris-latency-mobility.

References

[1] E. Basar, M. D. Renzo, J. D. Rosny, M. Debbah, M.-S. Alouini, and R. Zhang, "Wireless communications through reconfigurable intelligent surfaces," *IEEE Access*, vol. 7, pp. 116 753–116 773, 2019.
[2] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Transactions on Wireless Communications*, vol. 18, no. 11, pp. 5394–5409, 2019.

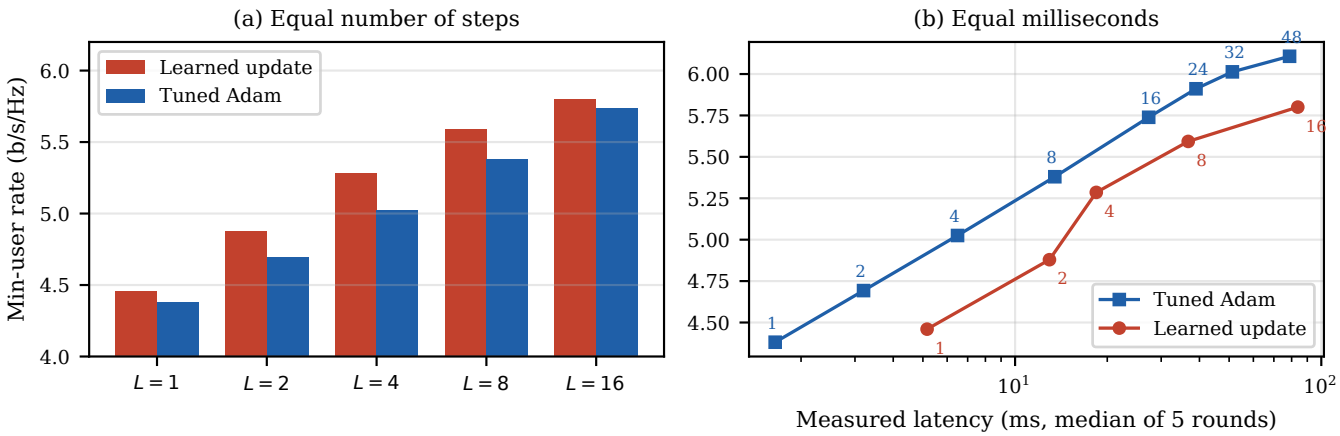


Fig. 5. Learned update rule versus tuned Adam (static test set, 600 scenarios, $t = 0$ rates, one CPU thread). (a) At equal steps the learned rule is ahead at every budget. (b) On the measured latency axis (labels: number of steps L), every learned configuration lies below and to the right of the Adam curve: it is dominated.

[3] M. D. Renzo, A. Zappone, M. Debbah, M.-S. Alouini, C. Yuen, J. de Rosny, and S. Tretyakov, "Smart radio environments empowered by reconfigurable intelligent surfaces: How it works, state of research, and the road ahead," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 11, pp. 2450–2525, 2020.

[4] Q. Wu, S. Zhang, B. Zheng, C. You, and R. Zhang, "Intelligent reflecting surface-aided wireless communications: A tutorial," *IEEE Transactions on Communications*, vol. 69, no. 5, pp. 3313–3351, 2021.

[5] L. Dai, B. Wang, M. Wang, X. Yang, J. Tan, S. Bi, S. Xu, F. Yang, Z. Chen, M. D. Renzo, C.-B. Chae, and L. Hanzo, "Reconfigurable intelligent surface-based wireless communications: Antenna design, prototyping, and experimental results," *IEEE Access*, vol. 8, pp. 45 913–45 923, 2020.

[6] X. Pei, H. Yin, L. Tan, L. Cao, Z. Li, K. Wang, K. Zhang, and E. Björnson, "RIS-aided wireless communications: Prototyping, adaptive beamforming, and indoor/outdoor field trials," *IEEE Transactions on Communications*, vol. 69, no. 12, pp. 8627–8640, 2021.

[7] S. Abeywickrama, R. Zhang, Q. Wu, and C. Yuen, "Intelligent reflecting surface: Practical phase shift model and beamforming optimization," *IEEE Transactions on Communications*, vol. 68, no. 9, pp. 5849–5863, 2020.

[8] C. Pan, H. Ren, K. Wang, W. Xu, M. Elkashlan, A. Nallanathan, and L. Hanzo, "Multicell MIMO communications relying on intelligent reflecting surfaces," *IEEE Transactions on Wireless Communications*, vol. 19, no. 8, pp. 5218–5233, 2020.

[9] H. Guo, Y.-C. Liang, J. Chen, and E. G. Larsson, "Weighted sum-rate maximization for reconfigurable intelligent surface aided wireless networks," *IEEE Transactions on Wireless Communications*, vol. 19, no. 5, pp. 3064–3076, 2020.

[10] H. Choi, A. L. Swindlehurst, and J. Choi, "WMMSE-based rate maximization for RIS-assisted MU-MIMO systems," *IEEE Transactions on Communications*, vol. 72, no. 8, pp. 5194–5208, 2024.

[11] C. Huang, R. Mo, and C. Yuen, "Reconfigurable intelligent surface assisted multiuser MISO systems exploiting deep reinforcement learning," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 8, pp. 1839–1850, 2020.

[12] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project (3GPP), Technical Report TR 38.901, V16.1.0 (Release 16), Dec. 2019.

[13] X. Yu, D. Xu, and R. Schober, "MISO wireless communication systems via intelligent reflecting surfaces," in *Proc. IEEE/CIC International Conference on Communications in China (ICCC)*, 2019, pp. 735–740.

[14] Q. Shi, M. Razaviyayn, Z.-Q. Luo, and C. He, "An iteratively weighted MMSE approach to distributed sum-utility maximization for a MIMO interfering broadcast channel," *IEEE Transactions on Signal Processing*, vol. 59, no. 9, pp. 4331–4340, 2011.

[15] T. Jiang, H. V. Cheng, and W. Yu, "Learning to reflect and to beamform for intelligent reflecting surface with implicit channel estimation," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 7, pp. 1931–1945, 2021.

[16] V. Monga, Y. Li, and Y. C. Eldar, "Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing," *IEEE Signal Processing Magazine*, vol. 38, no. 2, pp. 18–44, 2021.

[17] Y. Liu, Q. Hu, Y. Cai, G. Yu, and G. Y. Li, "Deep-unfolding beamforming for intelligent reflecting surface assisted full-duplex systems," *IEEE Transactions on Wireless Communications*, vol. 21, no. 7, pp. 4784–4800, 2022.

[18] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Transactions on Signal Processing*, vol. 67, no. 10, pp. 2554–2564, 2019.

[19] Q. Hu, Y. Cai, Q. Shi, K. Xu, G. Yu, and Z. Ding, "Iterative algorithm induced deep-unfolding neural networks: Precoding design for multiuser MIMO systems," *IEEE Transactions on Wireless Communications*, vol. 20, no. 2, pp. 1394–1410, 2021.

[20] L. Pellaco, M. Bengtsson, and J. Jaldén, "Matrix-inverse-free deep unfolding of the weighted MMSE beamforming algorithm," *IEEE Open Journal of the Communications Society*, vol. 3, pp. 65–81, 2022.

[21] H. Sun, X. Chen, Q. Shi, M. Hong, X. Fu, and N. D. Sidiropoulos, "Learning to optimize: Training deep neural networks for interference management," *IEEE Transactions on Signal Processing*, vol. 66, no. 20, pp. 5438–5453, 2018.

[22] S. Savaşçı Şen, A. Çalhan, M. Cicioğlu, and O. Demiryürek, "Overcoming the phase discontinuity problem in RIS optimization: A robust and scalable phase-aware deep regression framework via hybrid CNN–LSTM architecture," *Advanced Intelligent Systems*, vol. 8, no. 9, p. e70460, 2026.

[23] Z. Huang, B. Zheng, and R. Zhang, "Transforming fading channel from fast to slow: Intelligent refracting surface aided high-mobility communication," *IEEE Transactions on Wireless Communications*, vol. 21, no. 7, pp. 4989–5003, 2022.

[24] A. Papazafeiropoulos, I. Krikidis, and P. Kourtessis, "Impact of channel aging on reconfigurable intelligent surface aided massive MIMO systems with statistical CSI," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 1, pp. 689–703, 2023.

[25] Y. Zhang, J. Zhang, M. D. Renzo, H. Xiao, and B. Ai, "Reconfigurable intelligent surfaces with outdated channel state information: Centralized vs. distributed deployments," *IEEE Transactions on Communications*, vol. 70, no. 4, pp. 2742–2756, 2022.

[26] C. Liu, W. Yuan, Z. Wei, X. Liu, and D. W. K. Ng, "Location-aware predictive beamforming for UAV communications: A deep learning approach," *IEEE Wireless Communications Letters*, vol. 10, no. 3, pp. 668–672, 2021.

[27] H. Han, T. Jiang, and W. Yu, "Active sensing for multiuser beam tracking with reconfigurable intelligent surface," *IEEE Transactions on Wireless Communications*, vol. 24, no. 1, pp. 540–554, 2025.

[28] K. T. Truong and R. W. Heath, "Effects of channel aging in massive MIMO systems," *Journal of Communications and Networks*, vol. 15, no. 4, pp. 338–351, 2013.

[29] R. Chopra, C. R. Murthy, H. A. Suraweera, and E. G. Larsson, "Performance analysis of FDD massive MIMO systems under channel aging," *IEEE Transactions on Wireless Communications*, vol. 17, no. 2, pp. 1094–1108, 2018.

[30] N. Jindal, "MIMO broadcast channels with finite-rate feedback," *IEEE Transactions on Information Theory*, vol. 52, no. 11, pp. 5045–5060, 2006.
[31] M.-M. Zhao, Q. Wu, M.-J. Zhao, and R. Zhang, "Intelligent reflecting surface enhanced wireless networks: Two-timescale beamforming optimization," *IEEE Transactions on Wireless Communications*, vol. 20, no. 1, pp. 2–17, 2021.
[32] Y. Cao, T. Lv, and W. Ni, "Two-timescale optimization for intelligent reflecting surface-assisted MIMO transmission in fast-changing channels," *IEEE Transactions on Wireless Communications*, vol. 21, no. 12, pp. 10 424–10 437, 2022.
[33] Q. Wu and R. Zhang, "Beamforming optimization for wireless network aided by intelligent reflecting surface with discrete phase shifts," *IEEE Transactions on Communications*, vol. 68, no. 3, pp. 1838–1851, 2020.
[34] Y. Zhao, Y. Feng, A. M. Ismail, Z. Wang, Y. L. Guan, Y. Guo, and C. Yuen, "2-bit RIS prototyping enhancing rapid-response space-time wavefront manipulation for wireless communication: Experimental studies," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 4885–4901, 2024.
[35] C. B. Peel, B. M. Hochwald, and A. L. Swindlehurst, "A vector-perturbation technique for near-capacity multiantenna multiuser communication—Part I: Channel inversion and regularization," *IEEE Transactions on Communications*, vol. 53, no. 1, pp. 195–202, 2005.
[36] Q.-U.-A. Nadeem, A. Kammoun, A. Chaaban, M. Debbah, and M.-S. Alouini, "Asymptotic max-min SINR analysis of reconfigurable intelligent surface assisted MISO systems," *IEEE Transactions on Wireless Communications*, vol. 19, no. 12, pp. 7748–7764, 2020.
[37] M. Gudmundson, "Correlation model for shadow fading in mobile radio systems," *Electronics Letters*, vol. 27, no. 23, pp. 2145–2146, 1991.
[38] W. Tang, M. Z. Chen, X. Chen, J. Y. Dai, Y. Han, M. D. Renzo, Y. Zeng, S. Jin, Q. Cheng, and T. J. Cui, "Wireless communications with reconfigurable intelligent surface: Path loss modeling and experimental measurement," *IEEE Transactions on Wireless Communications*, vol. 20, no. 1, pp. 421–439, 2021.
[39] 3GPP, "Study on scenarios and requirements for next generation access technologies," 3rd Generation Partnership Project (3GPP), Technical Report TR 38.913, V16.0.0 (Release 16), Jul. 2020.
[40] ITU-R, "Minimum requirements related to technical performance for IMT-2020 radio interface(s)," International Telecommunication Union, Report ITU-R M.2410-0, Nov. 2017.
[41] R. H. Clarke, "A statistical theory of mobile-radio reception," *Bell System Technical Journal*, vol. 47, no. 6, pp. 957–1000, 1968.
[42] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Prentice Hall, 2002.
[43] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. International Conference on Learning Representations (ICLR)*, 2015, arXiv:1412.6980.
[44] L. Armijo, "Minimization of functions having Lipschitz continuous first partial derivatives," *Pacific Journal of Mathematics*, vol. 16, no. 1, pp. 1–3, 1966.
[45] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.
[46] B. Zheng, C. You, W. Mei, and R. Zhang, "A survey on channel estimation and practical passive beamforming design for intelligent reflecting surface aided wireless communications," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 1035–1071, 2022.
[47] Z. Wang, L. Liu, and S. Cui, "Channel estimation for intelligent reflecting surface assisted multiuser communications: Framework, algorithms, and analysis," *IEEE Transactions on Wireless Communications*, vol. 19, no. 10, pp. 6607–6620, 2020.
[48] 3GPP, "NR; physical channels and modulation," 3rd Generation Partnership Project (3GPP), Technical Specification TS 38.211 (Release 16), 2020.